The University Of Queensland
AUSTRALIA
CREATE CHANGE

Studio 2 – Interaction Design
DECO2850 / DECO7285

Week 10: User Evaluation

Announcements

- Assessment 1 grades are announced
- Assessment 4 [Post grad] announced
- Draft Assessment 3 planning from next week

Why are we doing this?

Steps:

1. Form a Team and Define a Project
   - No technology/app at this stage
2. Plan a study to investigate the problem further
   - E.g. “How do UQ’s International students manage their diet?”
     - Develop a suit of methods (interviews, observations, etc.) to find answer to the question
     - Engage with potential users and collect data
3. Analyse data and develop initial ideas for d
4. Develop prototype – Iteration 1
5. Engage with same users and trial/ evaluate/ assess the prototype
6. Develop final prototype – Iteration 2

Week	Topic	Studio Activity
1. 30/07/26	Introduction	Introduction to the Course, Class overview, organisation, assessments, and project selection. Design Exercise
2. 06/08/26	Team formation	Speed Dating / Project Management Studio: Project idea brainstorming
3. 13/08/26	Understanding People	Methods: Interviews, Ethnography, Object-based Methods, Online Ethnography, etc. Studio
4. 20/08/26	Ethics & Values	Ethics, Values and Technology Studio
5. 27/08/26	Data Analysis	Qualitative Data Analysis Activity: Data Analysis
6. 03/09/26	Prototyping & Interactions	Prototyping Methods, API & Tangibles Activity: Prototyping
7. 10/09/26	UX	User Experience Design Studio
8. 17/09/26	Assessment 2: Milestone 1	Design Exhibit 1
9 24/09/26	User Enactment	Wizard of Oz, Role Playing, Bodystorming, Simulated Environments Activity: User Enactment
Semester Break		
10. 08/10/26	Evaluation	Evaluation methods Studio
11. 15/10/26	Field Trials	Field trial methods Studio
12. 22/10/26	Prep for Exhibit	Studio
13. 29/10/26	Assessment 2: Milestone 2	Design Exhibit 2
14.	SWOT VAC	
15.	Assessment 2: Milestone 3 13/11/2026, 3pm Assessment 3: Interviews between 9/11/2026 to 13/11/	
16.	Assessment 4 [PG only]: 16/11/2026, 3pm	

Today’s Topic

- 10:00 to 10:30 | Assessments
- 10:30 to 11:00 | Intro to User Evaluations
- 11:00 to 13:50 | Studio: Plan your evaluations

Outcome:

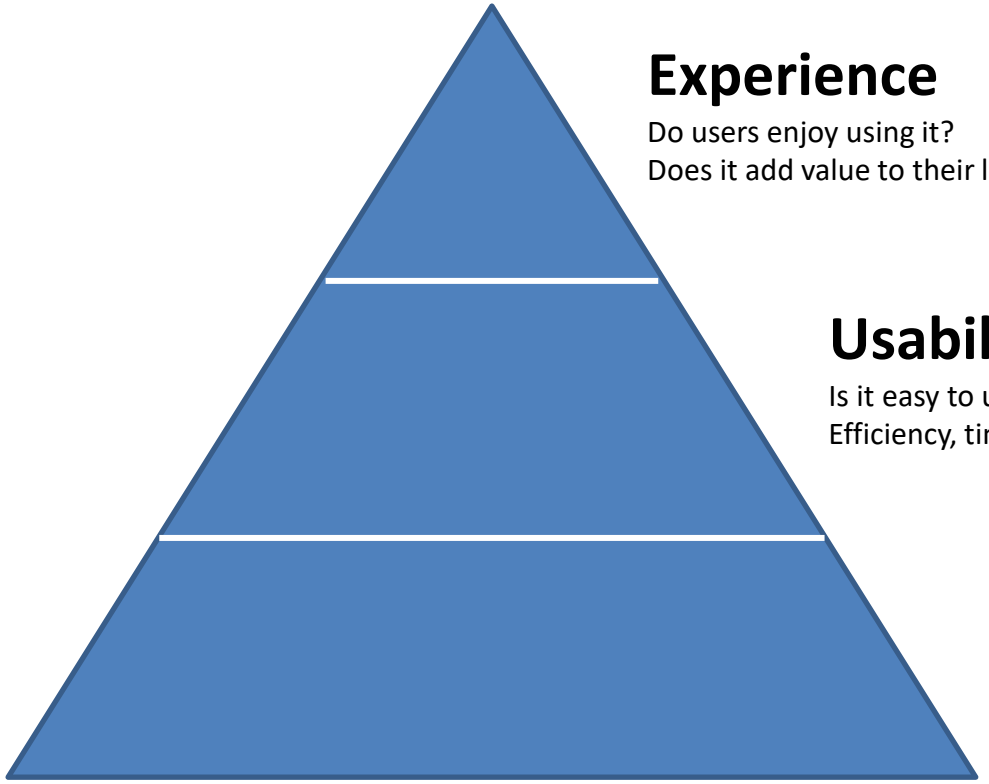
Be able to run use evaluation for your projects.

Evaluation in HCI

- Aim:
  - To assess whether the designed system supports the needs of users and to study how they feel using it.
- Occurs throughout the design life cycle
  - results feeding back into modifications to the design.
- Prototyping and Evaluation go hand-in-hand

Evaluation

What



Experience

Do users enjoy using it?
Does it add value to their lives?

Usability

Is it easy to use?
Efficiency, timeliness, etc.

Functionality

Does it work?
Does it do what it is supposed to do?

Evaluation Methods

1. Think-aloud protocols [Lab]
2. Audio/Video Narration [In-situ]
3. Experience Sampling [In-situ]
4. Day Reconstruction Method [In-situ]

1. Think Aloud

- Users are observed performing specific tasks
- Users are asked to describe what they are doing and why, what they think is happening, etc.
- Proceedings are recorded for analysis
- Advantages
  - simplicity - requires little expertise
  - can provide useful insight
  - can show how system is actually use
- Disadvantages
  - subjective
  - selective

1. Think Aloud





2. Audio/Video Narration

- Let the users verbally/visually report their experiences in a free story format.
- **Description**
  - The users tell a story of their experiences with a prototype; and their story is audio recorded. The task assignment can include some topics or questions to support the users.
- **Strengths**
  - The method will give the most important experiences that the users have with the product
- **Weaknesses**
  - Some users may not be comfortable telling about their experiences and the story might be very short.
  - The stories need to be transcribed for later analysis

2. Video Narration



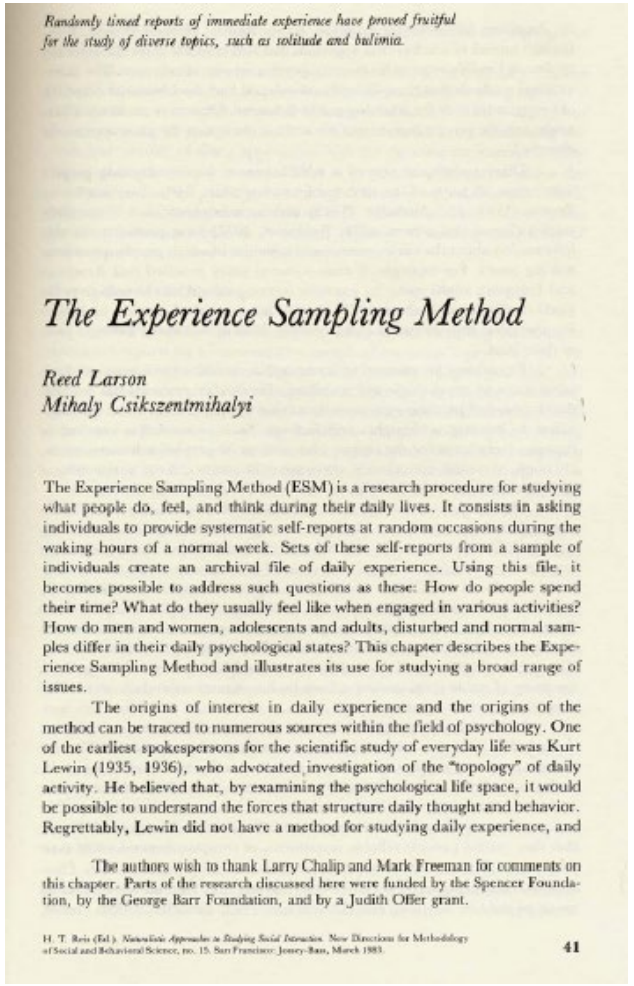


[Camera pointed at Ecosphere, close up, pointing with her free hand; describing enthusiastically]:"*This is the good thing so you can actually come out and you can analyse where it is that you've got your peak usage, so that will be the hair dryer drying my hair and that would be the blender. The pink one will be using the blender*" (Single occupant: Family 2).

Snow, S., Vyas, D. and Brereton, M. (2015). When an eco-feedback system joins the family. Personal & Ubiquitous Computing.

3. Experience Sampling Method (ESM)

- Larson / Csikszentmihalyi [1983]
- ESM is designed to repeatedly request people to document and report their thoughts, feelings and actions outside the laboratory, within the context of their everyday life.
- Process for studying what people
  - Do
  - Think
  - Feel
- Asking individuals to provide systematic self-reports
  - Random occasions
  - During specific hours



Randomly timed reports of immediate experience have proved fruitful for the study of diverse topics, such as solitude and bulimia.

The Experience Sampling Method

Reed Larson
Mihaly Csikszentmihalyi

The Experience Sampling Method (ESM) is a research procedure for studying what people do, feel, and think during their daily lives. It consists in asking individuals to provide systematic self-reports at random occasions during the waking hours of a normal week. Sets of these self-reports from a sample of individuals create an archival file of daily experience. Using this file, it becomes possible to address such questions as these: How do people spend their time? What do they usually feel like when engaged in various activities? How do men and women, adolescents and adults, disturbed and normal samples differ in their daily psychological states? This chapter describes the Experience Sampling Method and illustrates its use for studying a broad range of issues.

The origins of interest in daily experience and the origins of the method can be traced to numerous sources within the field of psychology. One of the earliest spokespersons for the scientific study of everyday life was Kurt Lewin (1935, 1936), who advocated investigation of the "topology" of daily activity. He believed that, by examining the psychological life space, it would be possible to understand the forces that structure daily thought and behavior. Regrettably, Lewin did not have a method for studying daily experience, and

The authors wish to thank Larry Chalip and Mark Freeman for comments on this chapter. Parts of the research discussed here were funded by the Spencer Foundation, by the George Barr Foundation, and by a Judith Offer grant.

H. T. Reis (Ed.). *Naturalistic Approaches to Studying Social Interaction*. New Directions for Methodology of Social and Behavioral Science, no. 15. San Francisco: Jossey-Bass, March 1983.

3. Experience Sampling Method (ESM)

- Interval-contingent sampling
  - Sample on experiences at fixed times
  - Good for time series data
  - Typically, burdensome to subjects
    - They begin to expect prompts
- Event-Contingent sampling
  - Report on experiences based on event of interest
  - Subject must be “cognitively-engaged” into own actions

4. Day Reconstruction Method

Capturing in-situ data retrospectively

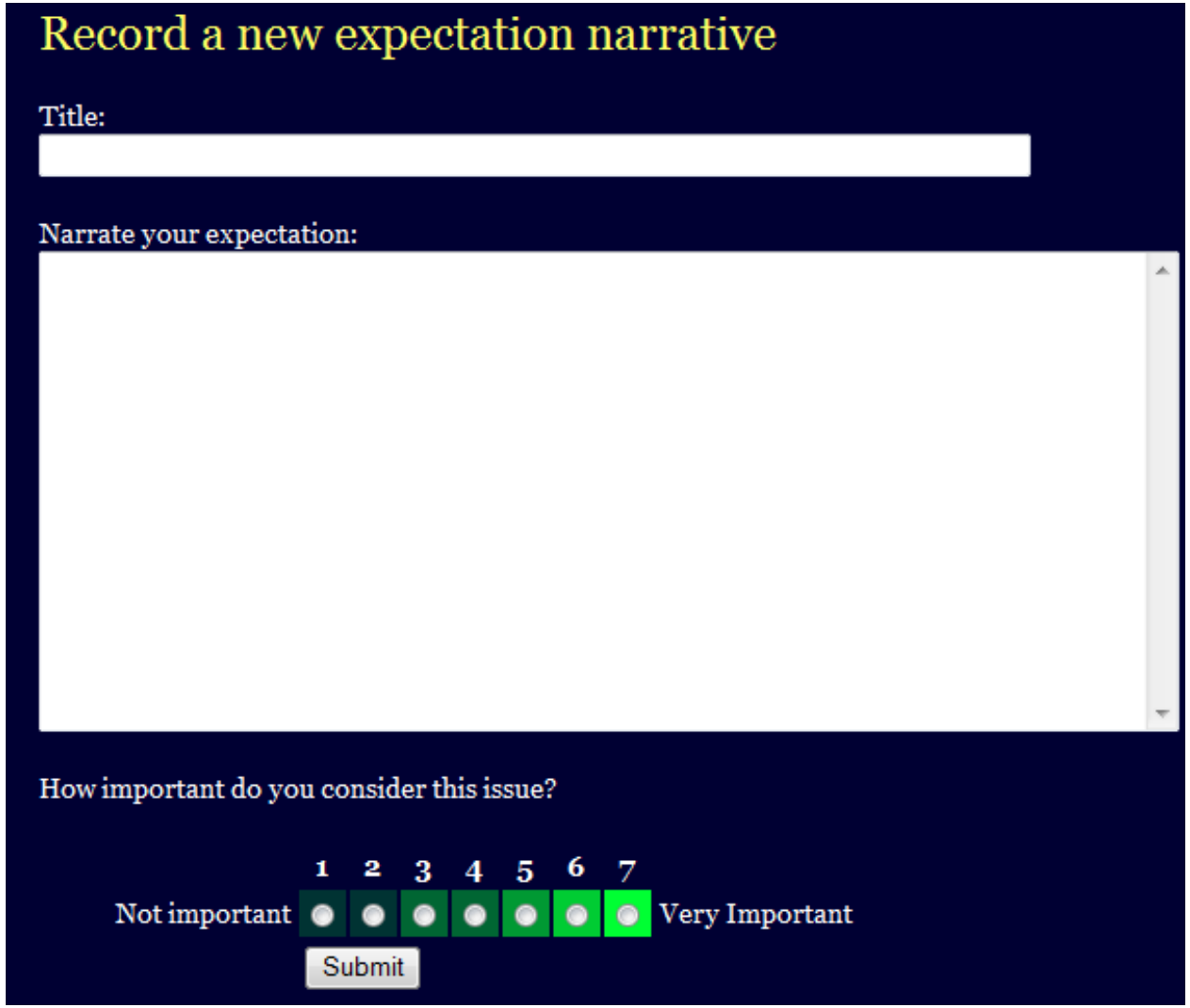
- Experience Sampling Method
  - High levels of participant burden
  - Not feasible to retrieve rich qualitative accounts
- How does DRM work?
  - Takes place at the end of the day
    - All experiences are listed in a chronological order
  - Every experience is recalled in relation to preceding ones

4. Day Reconstruction Method

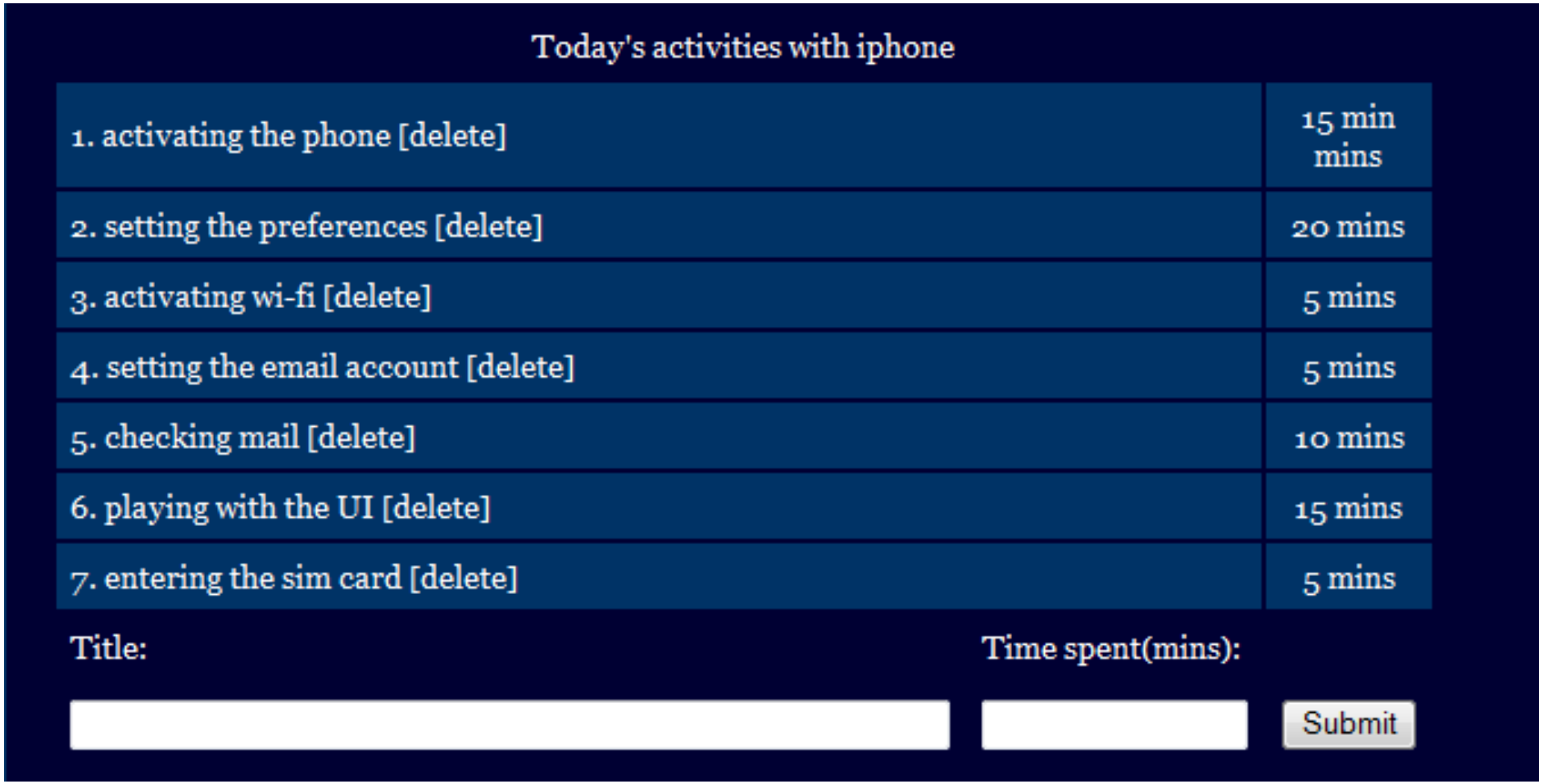
An Example [Karapanos, 2009]

- 4-week field study on “Adoption of iPhone”
- 6 participants captured their daily experiences at the end of each day by
  - Day reconstruction: participants listed all activities of the day that somehow related to their iPhone. A brief name and an estimation of time spent for each activity.
  - Experience narration: participants were asked to pick the three *most impactful,* either satisfying or dissatisfying, experiences of that day. For each of the three experiences, participants were asked to write a story that describes in detail the situation, their feelings and their momentary perceptions of the product.

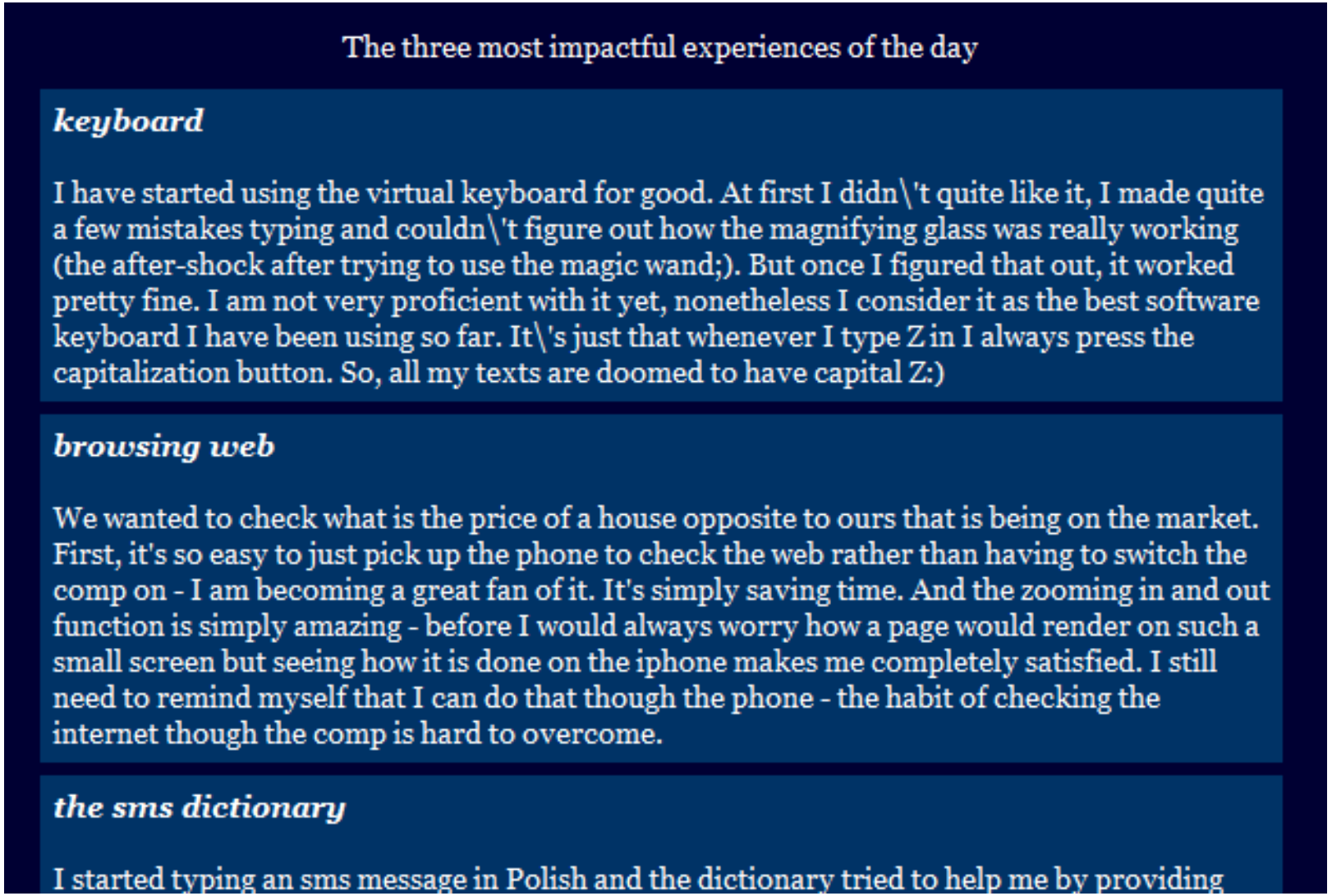
4. Day Reconstruction Method



4. Day Reconstruction Method



4. Day Reconstruction Method



4. Day Reconstruction Method

References:

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- Kahneman, Daniel, et al. "A survey method for characterizing daily life experience: The day reconstruction method." *Science*306.5702 (2004): 1776-1780.

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- Larson, R., & Csikszentmihalyi, M. (1983). "The experience sampling method". *New Directions for Methodology of Social and Behavioral Science*, 15, 41-56.
- “Interaction Design”. By Preece, Rogers & Sharp.
- “Designing Interactive Systems”. By Benyon, Turner & Turner.
- “Human Computer Interaction”. By Dix, Finley, Abowd.

Evaluation Activity

Steps:

1. Review evaluation methods
2. Select a method or a suite of methods
3. Use personas, or develop a new one

4. Plan your evaluation study
   - I.e. What questions will you ask…

Role Play of Evaluation

- Personas from each table will be shared with other teams
  - Tutors will help you
- One member from each team will volunteer to be a “user” for another team
  - They will review the persona and ‘pretend’ to be a potential user
- Each team will run their evaluation method – involving a “participant” (from another team)
- Summarize your evaluation findings


COFFEE
BREAK